Shortcomings of Computer Chat: Critical Utterances

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Abstract

It has been said that there is something deeply touching about creating something and then having a *chat* with it. Machines in general, and more specifically computers, are not the exception to that tendency. However, in spite of the accelerated advances in artificial intelligence (AI), the barrier of computer 'understanding' and naturally communicating in natural language has not been completely pulled down. In this paper we review the computer's shortcomings when having a chat with a user and propose our idea of the creation of critical utterances in order to have a better simulation of the human conversational behavior.

Keywords: natural language processing, human-computer interaction, critical utterances.

1 Introduction

Machine conversation with humans, as an area of research and development, has made exponential advances nowadays. However, in spite of the accelerated advances in artificial intelligence (AI), the barrier of computer 'understanding' and communicating in natural language has not been completely pulled down. On the other hand, through the application of AI techniques the simulation of the computer understanding of natural language is well-known. In this regard, excellent examples are the non-specific-goal dialogue[1] agents called *chatbots*.

A chatbot, or chat robot, is defined as a computer program that simulates human conversation through Artificial Intelligence (AI). In fact, ELIZA [1], pattern-matching natural language program whose goal is to assume the role of a Rogerian therapist during its dialogue with the user, has been considered the first created chatbot. Previous research has approached the problem of modelling computer dialogue, in fact three main approaches have been proposed [2]: dialogue grammars approach, plan-based approach, and joint action or collaborative approach. All of those approaches focused on creating a dialogue in order to achieve certain specific goal, it is to say, domain specific dialogue approaches. On the other hand, however, when it comes to *non-specific-goal* dialogue, it is remarkable the lack of research. In general, the dialogue modeling of a chatbot is based on certain words combinations that it finds in a phrase given by the user, and in the case of sophisticated chatbots, like ALICE [2] [3], AIML[3] is used to transcribe pattern-response relations. Chatbots are wide spread on the Web, being applied for electronic commerce customer service. A chatbot application to question answering has been proposed by C.S. Montero and K. Araki [4]. However, it is highly improbable to create all the possible patterns-response that could appear during a dialogue, a shortcoming that makes the chatbot to easily lose the conversation flow.

By analyzing the structure of human-human chats we realized that the dialogue could be interpreted as a critically self-organized system [5], where a single utterance could contribute to the continuation of the dialogue flow or could crash the interaction. In this paper we compare the structure of human-human chat with the structure of human-computer chat, becoming clear the lack of criticality of the latest, being this the main shortcoming when the computer is simulating human-human conversational behavior. In order to overcome this shortcoming, discovering what are the more likely flows a conversation could have, and then creating the appropriated utterances based on the obtained knowledge, we believe a smooth, more human-like machine chat could be achieved. Critical utterances in a conversation - utterances that could change or completely crash the flow of the dialogue - could emerge by applying Chance Discovery techniques [6].

[1] Open domain dialogue

[2] Artificial LInguistic Computer Entity

[3] Artificial Intelligence Markup Language, base on eXtended Markup Language, XML

2 Critical Utterances

A human chat section has been considered *critically self-organized* [5]. Self organized criticality states that large interactive systems naturally evolve toward a critical state in which a minor event can lead to catastrophe. Critically self-organized systems [7] as a scientific field of study is relatively new although the basic questions about how organization arises have been brought up since ancient times. In this regard it is been observed that many natural systems show organization, e.g., galaxies, planets, chemical compounds, cells, organisms and societies, therefore it can be said that complex phenomena observed everywhere indicate that nature operates at the self-organized critical state. Complex phenomena emerge when elements are interacting and connected into systems, as it is the case of the grains of sand in a sandpile [7]. The critical state is established entirely due to the dynamical interactions among individual elements of the system, implying self-organization. In the case of the sandpile paradigm, by adding grains of sand to the pile, eventually it reaches the state in which the addition of one single grain of sand can crash the whole pile, at this point it is said the pile has reached *the critical state*.

Considering a human-human chat section our system, we argue that the chat is a dynamic system whose environment is formed by both interlocutors utterances. The chat evolves from an initial flat state - *equilibrium* - to a state where a single utterance can completely change the direction of the conversation - catastrophe -, introducing a new topic, repeating the cycle again. This behavior has been called *Critically Self-Organized Chat* [5]. It could be argued that getting aware of the "critical behavior" of the human chat allows to create more human-like dialogue flows when modeling computer chat, therefore its importance.

In order to illustrate what was stated above, it is presented a fragment of a transcription of a real chat between two friends[4] (the speakers names have been omitted):

1A: Hello
1B: Hello A, ah
2A: Yes, B ah
2B: Ah
3A: Have you had your lunch
3B: No
4A: Uhm I haven't had my lunch also 4B: Uh busy uh
5A: Uhm busy playing with my nieces
5B: Both of them are in their own place
eh at your home now
3̈9̈A: You hear my little niece crying

[4] University of South California, Dialogue Diversity Corpus, Dialogue 91, http://www-rcf.usc.edu/~billmann/diversity/DDivers-site.htm

39B: Heh uh Uhm so are you spring cleaning already
40A: Ah intend to do it today lor
40B: Orh
41A: So I will be home whole day lah but
41B: Uhm
42A: Uhm you know like can't get started like that do a little bit then no don't want to do uhm then
42B: Uh huh uh huh
.....

In this chat, the utterances 1A to 4B form the flat initial state of the system: greetings and topic-introductory utterances. At this point there are not big changes in the chat, it could be said the system is in *equilibrium*. The next utterance - 5A - initiates the chat toward a determined topic: A's nieces. Although small "avalanches" may occur, it is to say, the focus of the topic may be slightly changed, this topic remains the same until utterance 39B "crashes" it (catastrophic avalanche) initiating a completely new topic: "laundry on weekends", and the cycle repeats itself. The utterance that leads to a catastrophe - changing the direction of the chat - has been called "critical utterance" [5].

3 Chance Discovery

Despite of being a new field of research and still in a early stage of development, chance discovery is being applied to a wide range of branches of science. Chance discovery, as a field of investigation itself, is devoted to analyze the occurrence of events that might have important impact in their happening area. In this regard, the word "chance" has been defined as the information about an event or situation significant for making decisions, being *chance discovery* the discovery of a chance, *not* by chance [8]. This chance has been considered a rare opportunity - or risk - whose significance has been unnoticed, which may lead to an unexpected benefit.

When it comes to the analysis of chat, it is precisely the dynamic structural behavior of human-human chat that allows the smooth transition from one topic to the other within a chat section. If we consider the time of transition between topics as potential chances, the analysis of critical utterances - when a topic changes - and their relationships becomes relevant. In this regard the application of chance discovery techniques, as data mining processing tool, are considered useful.

In order to analyze the dynamics of our system - a chat section - as to model how to identify potential chances - critical utterances or topic changing - a chance discovery graphical tool is needed. In this regard, the application of the *KeyGraph* becomes suitable. The *KeyGraph*, originally used for indexing documents [9], has developed as a data mining tool for extracting patterns of the appearance of chance events. As a data mining tool, it identifies relationship between terms in a document, particularly focusing on relationship of co-occurrence of both high probability and low probability events.

4 Analysis: Human-Human Chat vs. Human-Computer Chat

In order to improve human-computer chat a comparison with human-human chat was performed. The analysis was carried out similarly for the human-human chat section showed in Sect. 2 and the human-computer chat section (between a user and ALICE chatbot) as follow:

1) Each turn of the speakers - utterance - was considered one sentence.
2) Each sentence was segmented by words.
3) High frequency words were eliminated, i.e., I, you, are, my, and the like.
4) A vectorial representation of each sentence was created in order to find co-occurrence relation between them.
5) The co-occurrence document was analyzed using the *KeyGraph*

The graphic view of the human-human chat section is shown in Fig.1. In this figure the clusters of dots are representing the topics of the dialogue. The interconnection between the clusters are representing the transition between one topic and the next. It is noticeable the high interconnection between the clusters of the graph due to the dynamics of the chat flow, showing its criticality.

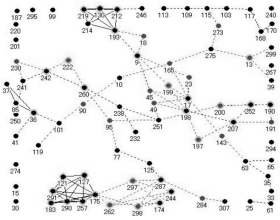


Fig. 1 Critical Chat: Human-Human Interaction

The graphic view of the human-computer chat section (48 turns of the interlocutors) is shown in Fig.2. It is remarkable the sparseness of the clusters of the graph, indicating a very rough transition between the topics of the dialogue. The lack of criticality in the human-computer chat section can be appreciated with the lack of interconnection of the graph.

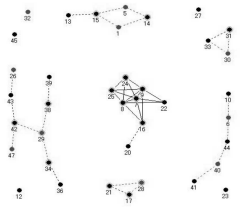


Fig. 2 Non-Critical Chat: Human-Computer Interation

5 Concluding Remarks and Future Work

In this paper a comparison between human-human chat and human-computer chat was preformed. The lack of criticallity of the human-computer chat could be identified as the main computer's shortcoming when interacting with the user, making the interaction vague and loosy. Based on this results the improvement of the chatbot database with critical utterances, as to shift smoothly from one topic to the next, remains as a near future work.

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